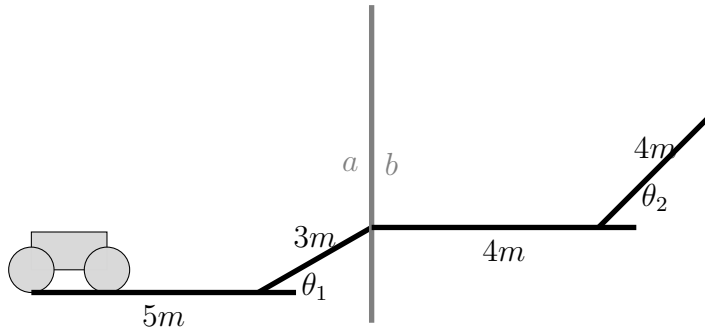


4. Below, a motorized cart travels across a road. The road has a coefficient of friction of $\mu = 0.5$. $\theta_1 = 30^\circ$ and $\theta_2 = 45^\circ$. The cart weighs 100 kg



- (a) Does the friction force do more work in part a or b of the road?
- (b) How much work does gravity do in part a and b ?
- (c) If it takes the cart 5 s to complete part a , and 7 s to complete part b , which part required the cart's motor to exert more power?